CS311

Introduction to Web Services Development

Complete Notes — Module 1 to Module 10

Virtual University of Pakistan

Prepared by:

BEYZA AYSUN

MODULE 1 — Object Oriented Paradigm

Concept	Meaning
Object	Real world entity
Class	Blueprint of objects
State	Properties
Behavior	Actions
Abstraction	Important cheezein dikhana
Inheritance	Properties inherit karna
Generalization	Common cheezein combine karna

Concept	Meaning
Specialization	General class se specific class banana
Extension	Existing class mein new features add karna
Restriction	Child class mein limitations lagana
Encapsulation	Data hide karna
Data Hiding	Internal details hide karna
Public	Sab access kar sakte
Private	Sirf class ke andar
Protected	Child classes access kar sakti hain

Concept	Meaning
Information Hiding	Important data hide karna
Single Inheritance	One parent → one child
Multilevel Inheritance	Grandparent → Parent → Child
Multiple Inheritance	Multiple parents → one child
Association	Objects ka relationship
One Way Association	Relation sirf ek direction mein
Two Way Association	Dono directions mein relation
Ternary Association	3 entities ka relation

Composition	Aggregation

Composition	Aggregation
Strong relationship	Weak relationship
Child independent nahi hota	Child independent hota
Parent delete → child bhi delete	Parent delete → child zinda
Example: House-Room	Example: Department-Teacher

Abstract Class	Interface
Some implementation ho sakti	Sirf method declarations
Inheritance use hota	Implements use hota
“is-a” relationship	Contract/service

Overloading	Overriding
Same class me hoti	Parent-child me hoti
Parameters different	Parameters same
Compile time	Run time
Example: add(int), add(double)	Boy.eat()

- Composition → Child parent ke bina nahi reh sakta
- Aggregation → Child independent hota
- Abstract Class → Blueprint jiska object nahi banta
- Concrete Class → Complete class jiska object banta
- Interface → Contract
- Overloading → Same method, different parameters
- Overriding → Child parent method ko replace karta hai

Concept	Meaning
Overloading	Same method, different parameters
Overriding	Child replaces parent method
Polymorphism	One interface, many forms
Dynamic Binding	Runtime pe method decide
Up-casting	Child object ko parent type banana

Up-casting	Down-casting
Child → Parent	Parent → Child

Up-casting	Down-casting
Automatic	Manual
Safe	Unsafe
Implicit	Explicit

MODULE 2 — Basics of Java Programming

Variable Types Comparison

Feature	Local	Instance	Static
Kahan banti	Method	Object	Class
Memory	Stack	Heap	Static memory
Default value	Nahi	Haan	Haan
Access modifier	Nahi	Haan	Haan
Lifetime	Method tak	Object tak	Program tak

Easy Real-Life Analogy

Local Variable

Exam hall me rough work.
Temporary.

Instance Variable

Har student ka:

- name
- roll number

Different.

Static Variable

School ka naam.
Sab students ke liye same.

Short Summary

JVM

- Java program ko run karti hai
- Bytecode execute karti hai
- Memory manage karti hai
- Garbage collection karti hai

Data Types

Primitive

Basic values:

- int
- char
- float

Non-Primitive

Objects/reference:

- String
- Array

Arithmetic Operators

Operator	Kaam
+	Add
-	Subtract
*	Multiply
/	Divide
%	Remainder
++	Increase by 1
--	Decrease by 1

MODULE 3 — Program Control Flow

Final Short Summary

IF

Decision lene ke liye.

```
if(condition)
```

IF-ELSE

True aur false dono cases handle karta hai.

```
if(condition)
else
```

FOR LOOP

Jab repetitions ka pata ho.

```
for()
```

WHILE LOOP

Jab tak condition true ho.

```
while()
```

DO-WHILE LOOP

Minimum 1 baar zarur execute hota hai.

```
do
{
}
while();
```

Topic	Meaning
Nested If	if ke andar if
Local Block	{ } ke andar code
Local Scope	Variable sirf block mein accessible
Semicolon	Statement end karta hai
Package	Classes organize karta hai
Class	Blueprint
Comments	Notes for programmer
main()	Program start point
println()	Output print karta hai

Topic	Meaning
JDK	Java tools package
Sentinel Loop	Unknown repetitions wali loop

Topic	Meaning
Sentinel Variable	Loop stop karne wala variable
Abstraction	Important cheezein dikhana
Encapsulation	Data hide karna
Class	Blueprint
Object	Class ka instance
new	Object create karta hai
Object Reference	Object ka address hold karta hai
Heap Memory	Object store hota hai
Stack Memory	Reference variable store hota hai

MODULE 4 & 5 — Using Objects / Primitive as Objects

Topic	Meaning
String	Characters ka sequence
Immutable	Change nahi hoti
String Pool	Memory save karta hai
length()	Characters count
charAt()	Specific character
concat()	Strings join
+ operator	Concatenation
equals()	Content compare
==	Reference compare
compareTo()	Alphabetical compare

Summary

Wrapper Class
Primitive ko object banati hai.

Autoboxing
Primitive → Object

Unboxing
Object → Primitive

Increment/Decrement
Value mein 1 add/minus.

For Loop
Known repetitions ke liye.

Nested Loop
Loop ke andar loop.

Array
Same datatype ki multiple values store karta hai.

Multidimensional Array
Array ke andar arrays.

Random Class
Random numbers generate karti hai.

MODULE 6 & 7 — Arrays / Classes

Final Summary

Class
Blueprint/template hoti hai.

Object
Class ka real instance.

Constructor
Object initialize karta hai.

new Operator
Memory allocate karta hai.

Pass by Value
Copy pass hoti hai.

Arrays
Reference type hote hain.

Multiple References
Do variables same object ko point kar sakte hain.

Default Values
Fields ko default values milti hain.

Local Variables
Default values nahi milti.

Local Variable	Field Variable
Method ke andar	Class ke andar
Temporary	Object ke sath rehta
Default value nahi	Default value milti
Sirf local scope	Global/class scope

Final Summary

Method
Code ka reusable block.

Access Modifiers
Accessibility control karte hain.

private
Sirf same class mein accessible.

public
Har jagah accessible.

Return Type
Method kya return karega.

void
Kuch return nahi karega.

Parameters
Input values.

Local Variables
Temporary variables inside method.

Field Variables
Class level variables jo sab methods use kar sakte hain.

Concept	Meaning
Mutator	Value change karta hai
Selector	Value read karta hai
Utility	Helper method
Static	Sab objects share karte hain
Instance	Har object ki apni copy
Formal Parameter	Method definition wali variable
Actual Parameter	Method call wali value
this	Current object
Pass-by-value	Copy pass hoti hai

Topic	Meaning
Scope	Variable kahan use ho sakta hai
Lifetime	Variable kitni der memory mein rehta hai
Local Variable	Method ke andar
Instance Variable	Object ke sath
Static Variable	Class ke sath
Constructor	Object initialize karta hai
Default Constructor	No parameter constructor
Garbage Collector	Unused objects remove karta hai

Topic	Meaning
Default Constructor	Automatic constructor
Parameterized Constructor	Values lene wala constructor
Abstraction	Important cheezen show karna
Block	{ } ke andar code
if	True par code chale
if-else	True/False dono cases handle

MODULE 8 — Object Communication

Topic	Meaning
Switch-case	Multiple conditions handle karna
For Loop	Fixed times loop chalana
While Loop	Condition true tak loop
Do-While Loop	Kam az kam 1 baar loop
Structured Programming	Organized aur easy code
Duplicate Code	Same code baar baar
Object Communication	Objects methods call karte hain
this	Current object ko refer karta hai
Reference Variable	Object ka address store karta hai

MODULE 9 — Modifiers

Topic	Meaning
Association	Do objects/classes ka relation
Aggregation	Weak relationship
Composition	Strong whole-part relationship
Encapsulation	Data hide aur secure karna
Abstraction	Important cheezen show karna
Public	Har jagah accessible
Protected	Same package + subclasses
Default	Sirf same package
Private	Sirf same class

Topic	Meaning
Public	Har jagah accessible
Protected	Same package + subclasses
Default	Sirf same package
Private	Sirf same class
Static	Class level variable/method
Final	Change ya override nahi ho sakta
Abstract	Incomplete class/method
Synchronized	One thread at a time
Native	Java ke bahar implement
Transient	Serialization mein save nahi hota

MODULE 10 — Exception Handling

Topic	Meaning
Volatile	Latest value sab threads ko mile
Exception	Runtime problem/error
Checked Exception	Compile time exception
Unchecked Exception	Runtime exception
Error	Serious system problem
try	Risky code block
catch	Exception handle karna
finally	Hamesha execute hota hai
throw	Manually exception throw
throws	Method declare karta hai exception

— End of Notes — BEYZA AYSUN —